Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/22/2015
PI Name:	Collins, Cynthia Ph.D.		
Project Title:	Gravitational effects on quorum sensing and biofilm formation		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Troy	**State:**	NY
Zip Code:	12180-3522	**Congressional District:**	21
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/01/2009	**End Date:**	05/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:			
Flight Assignment:	NOTE: Extended to 5/31/2014 per A. Chu/ARC and NSSC information (Ed., 5/22/13) NOTE: Extended to 5/31/2013 per A. Chu/ARC and NSSC information (Ed., 6/1/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Plawsky, Joel (Rensselaer Polytechnic Institute) Dordick, Jonathan (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNX09AI70G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	There is an urgent need to understand the effects of microgravity on the growth, cellular physiology, and cell-cell interactions in microbial biofilms. This information can then be used to curtail harmful activities of microbial consortia thriving as biofilms on the International Space Station and for the long-term success of human space exploration. Bacterial biofilms were abundant on the MIR space station and were responsible for increasing corrosion and blocking a water purification system. Health and safety hazards linked to the development of biofilms are also of great concern, including increased rates of infection due to the enhanced survival of organisms within biofilms and increased resilience of biofilm populations to antimicrobial compounds. Furthermore, experiments with Escherichia coli have indicated that biofilms formed under microgravity may have even higher levels of resistance to antimicrobials. We propose to study how gravity alters biofilm formation with the goal of developing new strategies to reduce their impact on the operation of spacecrafts and the health of their crew. We will focus on two model microorganisms that form biofilms both inside and outside of the human body: Pseudomonas aeruginosa and Staphylococcus aureus. These microbes can switch between benign and pathogenic interactions with humans and may be relevant to crew health during extended missions. We will examine how the three-dimensional structures of biofilms formed by these microbes and the requirements and regulation of cell-cell communication systems differ in microgravity compared to normal gravity. We will specifically focus on the role of cell-cell communication in mixed communities where ground-based studies		

	have indicated that inter-species communication can lead to increased virulence. We will employ newly developed, non-leaching, nanotechnology-based, antimicrobial surfaces to aid in assessing differences between biofilm formation and structure in normal and microgravity and to assess their potential to reduce the impact of biolfims in future spacecraft designs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	According to the Center for Disease Control (CDC), hospital-acquired infections are the fourth leading cause of death in the United States behind stroke, cancer, and heart disease. Furthermore, it is estimated that >65% of all bacterial infections are associated with biofilms. A greater understanding of biofilms is essential if we are to find effective methods to combat their formation. Furthermore, the low-shear conditions microbes experience in microgravity are similar to those found in the human body that are difficult to study. This work may provide new insights into the role of shear and other physical effects, such as convection, on biofilm formation.
Task Progress:	As a result of this work, we have provided important new information and insights into biofilm formation by two important bacteria during spaceflight. In 2013, we published a paper describing the results of our assays of biofilm formation by Pseudomonas aeruginosa during spaceflight in PLoS One. We observed that this organism both formed more biofilm during spaceflight relative to normal gravity controls and exhibited a novel structure. This work garnered significant press attention, including articles on space.com (http://www.space.com/) and Popular Science (<a target="_blank" href="http://www.popsci.com/science/article/2013-06/so-you-wanna-go-space-can-you-put-superpower-bacteria">http://www.popsci.com/</a). In 2013, we published a second paper in BMC Microbiology describing how planktonic P. aeruginosa cells are affected by the microgravity environment. Additional works describing mathematical modeling and S. aureus biofilm formation during spaceflight remain to be published.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Collins CH. "Biofilms grown by Pseudomonas aeruginosa during spaceflight show increased biomass and a column and canopy-shaped structure." Poster presentation at the 6th ASM (American Society for Microbiology) Conference on Biofilms, Miami, FL, September 29-October 4, 2012. 6th ASM (American Society for Microbiology) Conference on Biofilms, Miami, FL, September 29-October 4, 2012. , Oct-2012
Abstracts for Journals and Proceedings	Collins CH (presenter). Kim WS, Plawsky JL, Collins CH. "Biofilm Formation During Spaceflight: Insights From Mathematical Modeling of Growth in the Microgravity Environment." Oral presentation at the 2013 American Institute for Chemical Engineers Annual Meeting, San Francisco, CA, November 3-8, 2013. Conference abstract book. 2013 American Institute for Chemical Engineers Annual Meeting, San Francisco, CA, November 3--8, 2013. https://aiche.confex.com/aiche/2013/webprogram/Paper343838.html ; accessed 4/22/15. , Nov-2013
Articles in Peer-reviewed Journals	Kim W, Tengra FK, Shong J, Marchand N, Chan HK, Young Z, Pangule RC, Parra M, Dordick JS, Plawsky JL, Collins CH. "Effect of spaceflight on Pseudomonas aeruginosa final cell density is modulated by nutrient and oxygen availability." BMC Microbiol. 2013 Nov 6;13:241. http://dx.doi.org/10.1186/1471-2180-13-241 ; PubMed PMID: 24192060; PubMed Central PMCID: PMC4228280 , Nov-2013